

SÍNDROME COMPARTIMENTAL ABDOMINAL



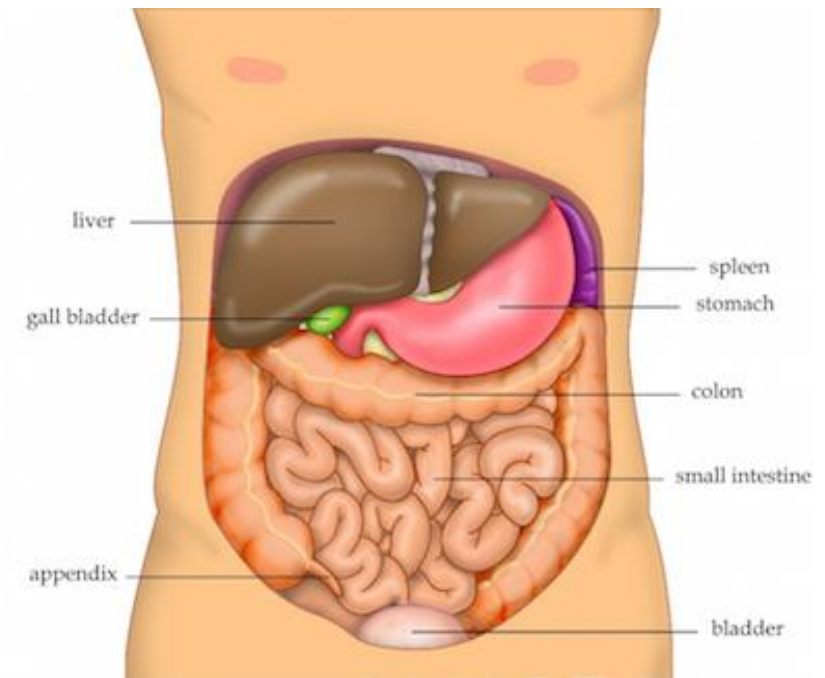
Bruna Scharanch

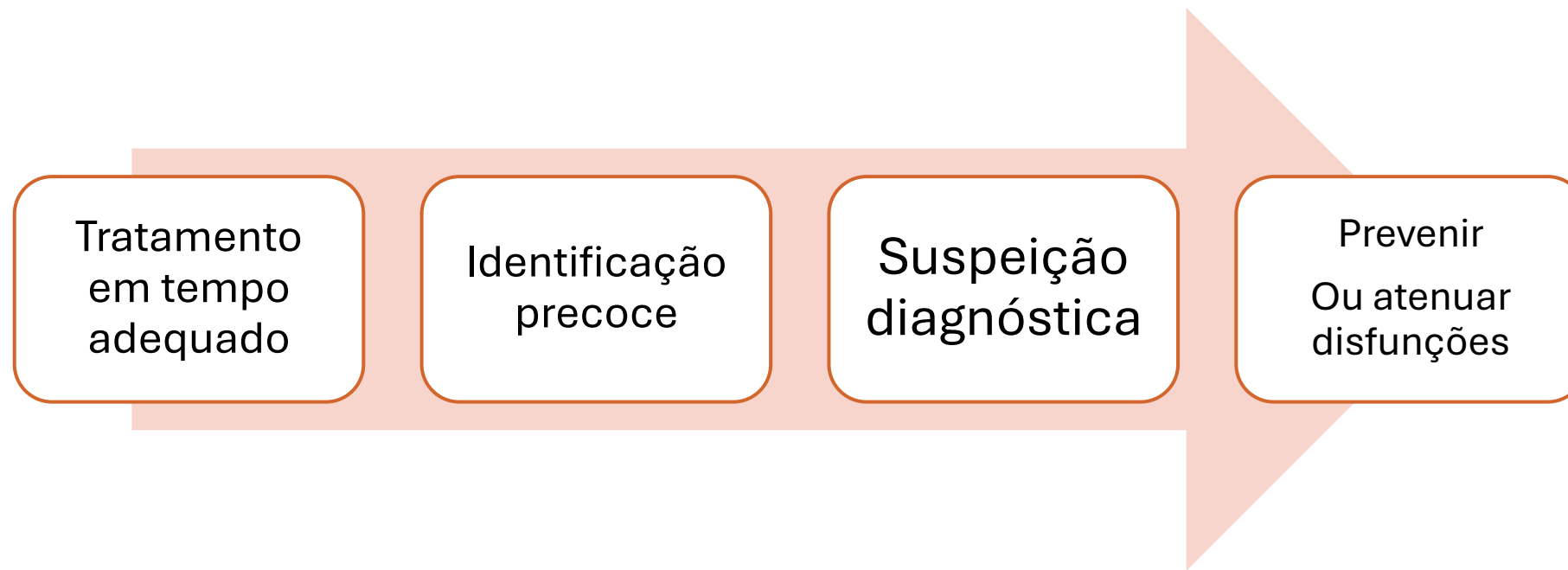
- Intensivista pelo HC FMUSP
- Clínica Médica e Graduação pela Universidade Federal do Triângulo Mineiro (UFTM)
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Introdução

- Condição grave
- Aumento tempo internação e morbimortalidade
- Subdiagnosticada (simular outras condições)
- Tratável
- Mesma PIA > Diferentes desfechos





Tempo de instalação:

- **Hiperaguda:** segundos – durante tosse, riso, esforço, evacuação, atividade física, espirro
- **Aguda:** horas – trauma, hemorragia – pode levar a SCA
- **Subaguda:** gradual dias - + comuns em doentes clínicos
- **Crônica:** meses (gravidez) ou anos (obesidade mórbida) – aumenta risco desenvolver SCA se sobreposição HIA aguda ou subaguda





- Aumento de um ou mais compartimentos intra cavitários



- Menor complacência da parede abdominal



- Comprometimento perfusão sanguínea dos órgãos abdominais



- Isquemia, acidose, dano tecidual, disfunção orgânica, óbito



Conceitos

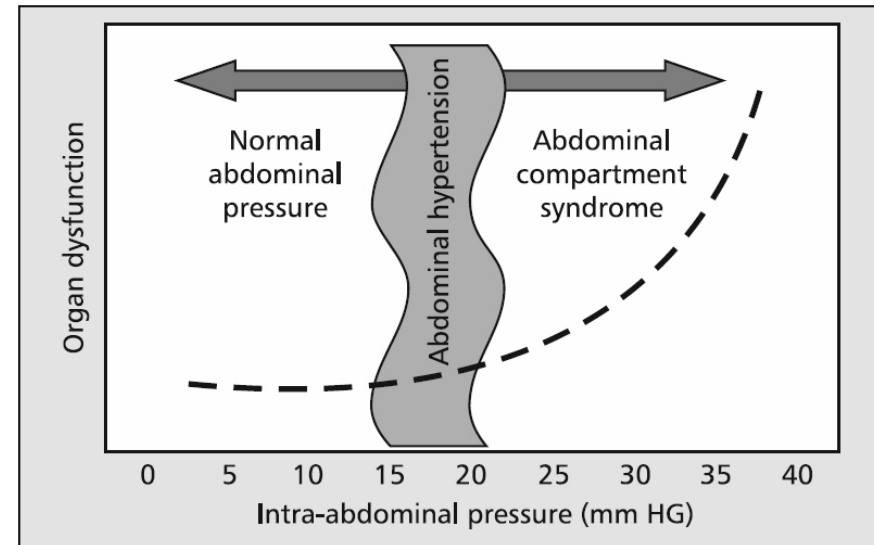
- PIA Normal → 5-7 mmHg (Obeso, Grávida 10-15**)
- HIA → PIA sustentada > 12 mmHg
- SCA → PIA sustentada > 20 mmHg + Disfunção Orgânica Nova



Conceitos

- HIA

- 12-15 mmHg → Grau 1
- 16-20 mmHg → Grau 2
- 21-25 mmHg → Grau 3
- >25mmHg → Grau 4





Epidemiologia

- Incidência Variável na população Crítica
- IAH 21-87%
- SCA 1-12%
- Mortalidade da SCA 40 a 100%
- Atualidade >>> Damage control, Balanço restritivo...



Risk factors for intra-abdominal hypertension and abdominal compartment syndrome among adult intensive care unit patients: a systematic review and meta-analysis

Jessalyn K Holodinsky¹, Derek J Roberts^{2*}, Chad G Ball³, Annika Reintam Blaser⁴, Joel Starkopf⁴, David A Zygun⁵, Henry Thomas Stelfox⁶, Manu L Malbrain⁷, Roman C Jaeschke⁸ and Andrew W Kirkpatrick⁹



Holodinsky et al. *Critical Care* 2013, **17**:R249
<http://ccforum.com/content/17/5/R249>

SCA:

- Cirurgia de emergência (<75min porta-cc)
- Pacientes “Graves”
- Acidose
- Oligúria
- Baixo DC
- Cristalóides >3L na Emergência
- Politransusão



Etiologia

- **Primária (lesão ou doença abdomino-pélvica)**

- Intervenções abdominais
- Trauma Abdominal
- Obstrução Intestinal
- Síndrome de Ogilvie
- Pancreatite
- Coleções intra-abdominais (Hemoperitônio/Ileo/Ascite)
- Abdominoplastia redutora – correção hérnias incisionais gigantes
- Transplantes – hepático, renal, intestinal e multivisceral

- **Secundária**

- Ressucitação Volêmica (principal causa)
- Grande Queimado - S.C. queimada > 30% - edema, redução complacência parede
- Sepses/SIRS



Incidence and prognosis of intraabdominal hypertension in a mixed population of critically ill patients: A multiple-center epidemiological study*

Manu L. N. G. Malbrain, MD; Davide Chiumello, MD; Paolo Pelosi, MD, PhD; David Bihari, MD; Richard Innes, MD; V. Marco Ranieri, MD, PhD; Monica Del Turco, MD; Alexander Wilmer, MD, PhD; Nicola Brienza, MD; Vincenzo Malcangi, MD; Jonathan Cohen, MD; Andre Japiassu, MD; Bart L. De Keulenaer, MD; Ronny Daelemans, MD; Luc Jacquet, MD, PhD; Pierre-François Laterre, MD, PhD; Günther Frank, MD; Paulo de Souza, MD; Bruno Cesana, MD; Luciano Gattinoni, MD, PhD

Design: Multiple-center, prospective epidemiologic study.

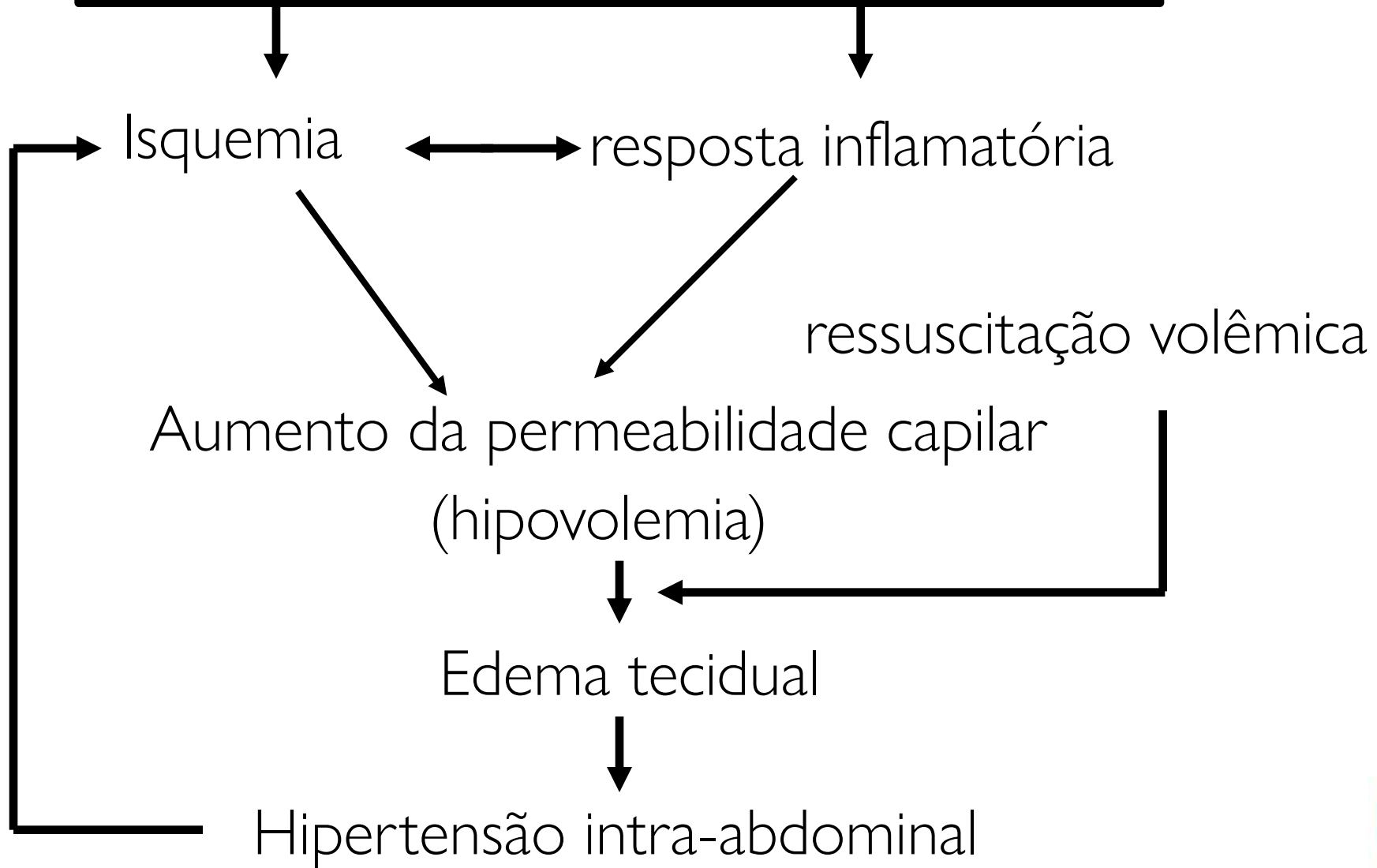
Setting: Fourteen intensive care units in six countries.

Patients: A total of 265 consecutive patients admitted for >24 hrs during the 4-wk study period.

	No IAH n = 180	IAH n = 85	p Value
Etiologic factors			
Abdominal surgery	42 (23.3)	38 (44.7)	<.0001
Hemoperitoneum	6 (3.3)	8 (9.4)	.04
Abdominal infection	23 (12.8)	13 (15.3)	NS
Fluid resuscitation	68 (37.8)	55 (64.7)	<.0001
Ileus	61 (33.9)	52 (61.2)	<.0001
>1	64 (35.6)	53 (62.4)	<.0001



Fisiopatologia da HIA/SCA



Consequências cardiovasculares :

Deslocamento do Diafragma



Compressão Cardíaca



Diminuição da complacência
e contratilidade ventricular



Diminuição do D.C.



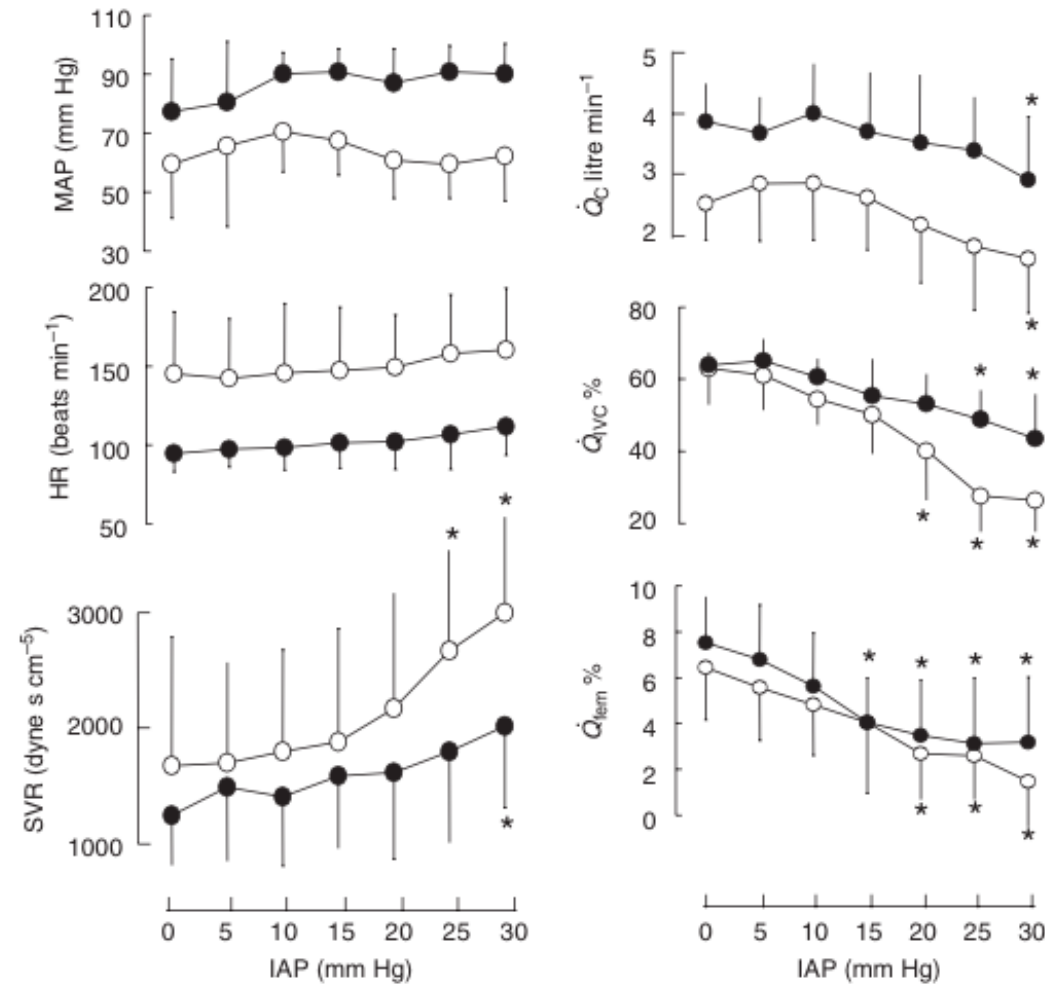
SCA



Diminuição do retorno venoso



Effects of increased intra-abdominal pressure on central circulation



Abdominal Perfusion Pressure: A Superior Parameter in the Assessment of Intra-abdominal Hypertension

Michael L. Cheatham, MD, Mark W. White, MD, Scott G. Sagraves, MD, Jeffrey L. Johnson, MD, and Ernest F. J. Block, MD

- $PPA = PAM - PIA$
- <55-60 menor sobrevida
- Não se sabe valor ideal
- Alvo > 60
- Quanto < PPA, > gravidade

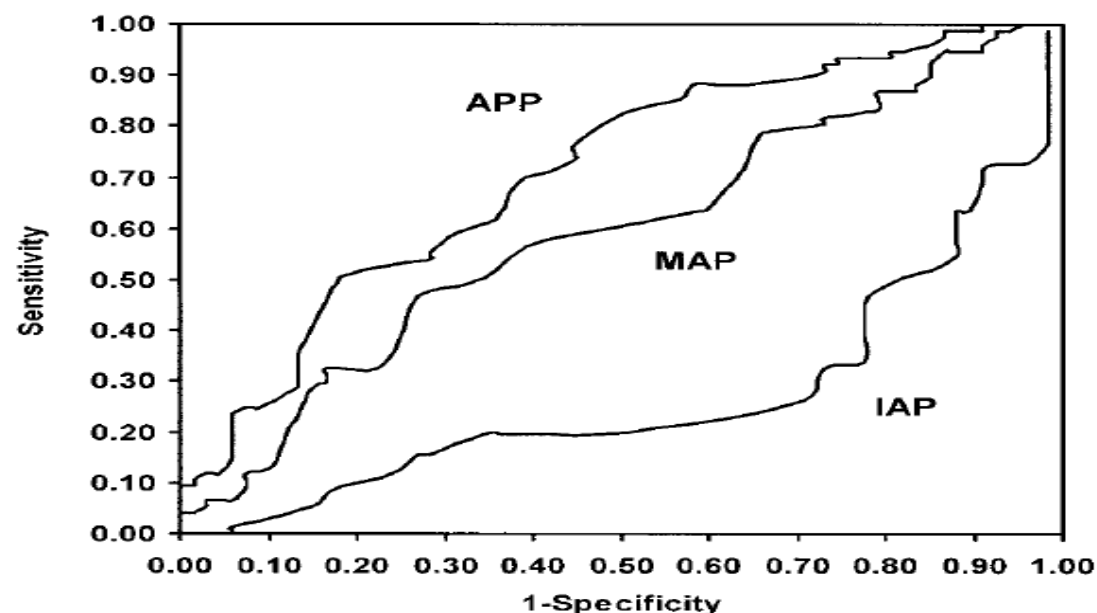
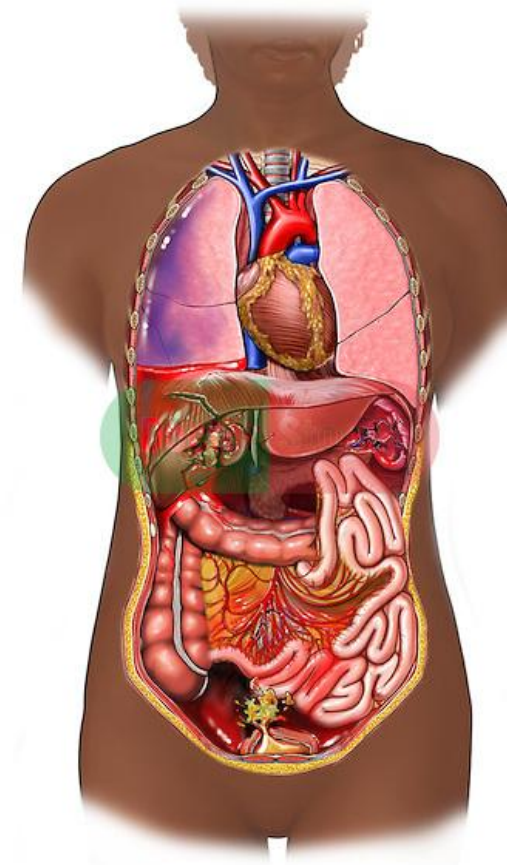


Fig. 1. ROC curve analysis of predictors of survival from IAH.



Consequências Pulmonares

- Complacência Pulmonar ↓
- Pressões inspiratórias ↑
(risco de barotrauma)
- Atelectasias + hipoxemia
- Aumento do PCO₂
- Volume corrente ↓
- Esforço inspiratório ↑



Effects of increased intra-abdominal pressure on central circulation

IAP (mm Hg)	0	5	10	15	20	25	30
Mean pulmonary arterial pressure (tm-mm Hg)							
Normovolaemic	26.7 (5.3)	29.3 (1.0)	32.4 (9.1)	34.7 (10.2)*	32.9 (10.6)	28.7 (6.8)	28.7 (8.7)
Hypovolaemic	20.2 (4.2)	20.7 (4.2)	22.5(5.0)	22.0 (5.5)	19.2 (6.7)	17.7 (7.2)	16.6 (6.1)
Pulmonary arterial occlusion pressure (tm-mm Hg)							
Normovolaemic	10.3 (2.6)	11.2 (3.1)	13.8 (3.3)	16.9 (4.9)*	13.9 (5.9)	12.1 (4.5)	11.2 (6.5)
Hypovolaemic	5.6 (4.0)	5.0 (4.3)	6.0 (3.5)	5.5 (3.5)	5.5 (5.9)	3.8 (4.3)	2.9 (3.1)
Left ventricular end-diastolic pressure (tm-mm Hg)							
Normovolaemic	12.0 (3.9)	14.5 (4.5)	15.6 (4.1)	16.6 (6.5)*	12.6 (8.5)	11.0 (9.2)	10.0 (10.4)
Hypovolaemic	6.1 (4.7)	8.2 (5.1)	7.1 (5.3)	5.9 (4.6)	4.9 (4.9)	3.6 (4.1)	2.2 (3.5)
Right atrial pressure (tm-mm Hg)							
Normovolaemic	9.2 (3.3)	10.8 (3.4)	13.8 (5.7)	14.3 (6.2)	13.6 (9.61)	12.6 (10.62)	14.3 (12.8)
Hypovolaemic	4.2 (3.7)	3.9 (4.1)	4.0 (4.2)	3.8 (4.4)	2.1 (4.4)	1.0 (3.6)	1.1 (3.2)
Peak inspiratory pressure (cm H ₂ O)							
Normovolaemic	20.1 (2.7)	21.7 (2.8)	25.9 (3.5)*	30.3 (4.5)*	32.7 (5.4)*	36.4 (7.4)*	38.9 (7.1)*
Hypovolaemic	20.1 (2.1)	21.2 (2.3)	25.0 (3.5)*	28.6 (4.3)*	32.3 (6.3)*	34.5 (6.2)*	36.4 (5.9)*
Mean airway pressure (cm H ₂ O)							
Normovolaemic	4.6 (1.1)	5.2 (1.3)	6.1 (1.9)	7.0 (1.8)*	7.6 (2.1)*	8.5 (2.1)*	9.3 (2.2)*
Hypovolaemic	4.7 (0.9)	5.1 (1.1)	6.0 (1.3)	6.7 (1.5)*	7.1 (1.5)*	7.6 (1.6)*	8.1 (1.8)*
End-expiratory pleural pressure (mm Hg)							
Normovolaemic	-0.9 (1.3)	-0.7 (1.7)	-0.2 (1.8)	-0.2 (1.9)	0.66 (1.8)	1.0 (1.9)	1.1 (1.8)
Hypovolaemic	-0.6 (2.5)	0.2 (2.9)	0.8 (2.6)	1.3 (2.6)	1.7 (3.2)	2.1 (3.6)	2.4 (3.4)



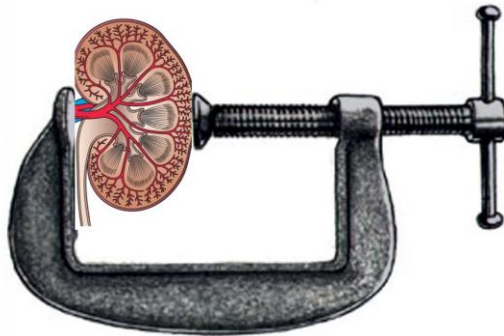
Baixo DC + Congestão Venosa + Vasoconstrição



Redução da perfusão



Diminuição do
Débito Urinário
IRA



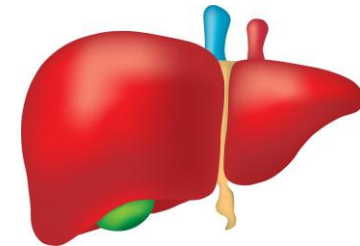
Neurológico:
Neuropatia
periférica,
parestesias, HIC,
redução drenagem
venosa cerebral,
hipoperfusão.
Sonolência
encefalopatia



Hematológico:
alteração
coagulação,
aumento risco
TVP e embolia



Isquemia hepática
Perda da atividade metabólica
Aumento lactato



Abdominal perfusion pressure is superior from intra-abdominal pressure to detect deterioration of renal perfusion in critically ill patients

Fethi Gül¹, İsmet Sayan¹, Umut Sabri Kasapoğlu¹, Derya Özer Erol¹,
Mustafa Kemal Arslantaş¹, İsmail Cinel¹, Zuhale Aykaç¹

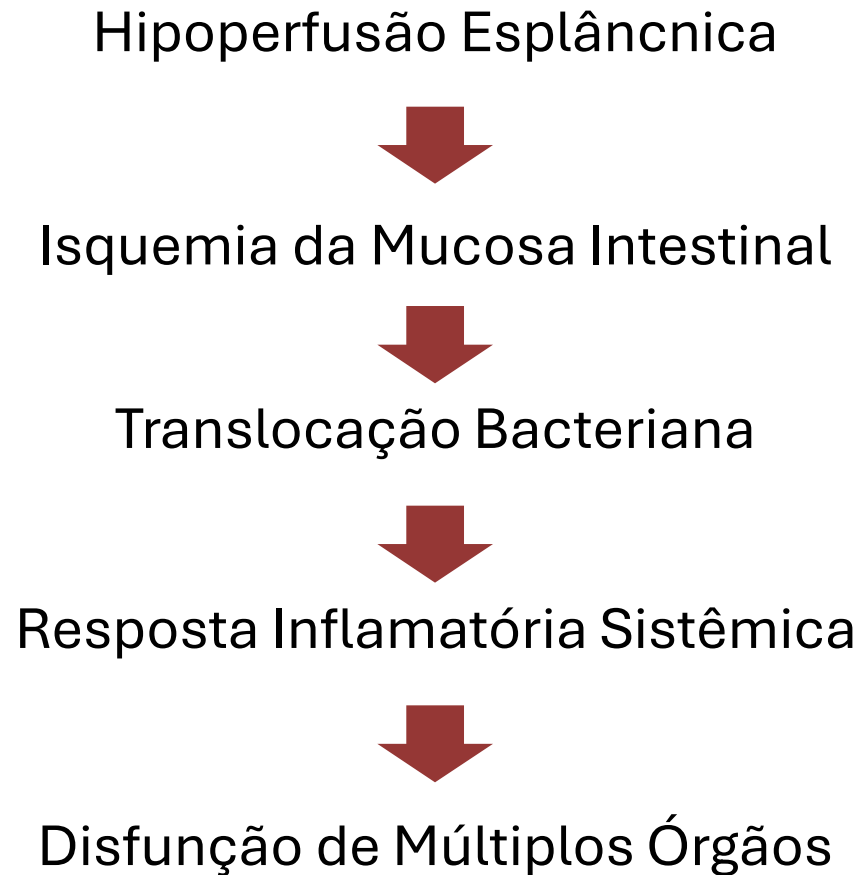
Methods: In this study, 38 patients

two groups according to renal dysfunction (Group I: RRI <0.72 vs Group II: RRI >0.72).

Results: The mean IAPs were not significant between the groups, 11.5±6.9 mm Hg in Group I (n=35) and 13.5±5.8 in Group II (n=33), respectively. APPs were statistically higher in Group I (81.2±13.6) than Group II (66.4±9.5) (p<0.001). The AUC for the association between APP at RRI >0.72 was 0.802 (p<0.001), with the APP ≤72 mmHg having a sensitivity of the 76% (95% CI 58-89%) and a specificity of 71% (95% CI 54-85%).



Consequências Gastrointestinais :



EFFECT OF INCREASED INTRA-ABDOMINAL PRESSURE ON MESENTERIC ARTERIAL AND INTESTINAL MUCOSAL BLOOD FLOW

Lawrence N. Diebel, MD, Scott A. Dulchavsky, MD, and Robert F. Wilson, MD

Table 2
Splanchnic hemodynamic measurements

IAP	SMA (mL/min)	SMA (% of Baseline)	MBF (% of Baseline)	pH _i
Baseline	128 ± 27	100	100	7.27 ± 0.10
10 mm Hg	135 ± 28	105 ± 22	81 ± 32	
20 mm Hg	94 ± 28*	73 ± 22*	61 ± 12*	7.16 ± 0.10*
30 mm Hg	61 ± 32†	48 ± 25†	45 ± 18†	
40 mm Hg	40 ± 19‡	31 ± 15‡	28 ± 8‡	6.98 ± 0.14†

* $p < 0.05$.

† $p < 0.01$.

‡ $p < 0.001$.

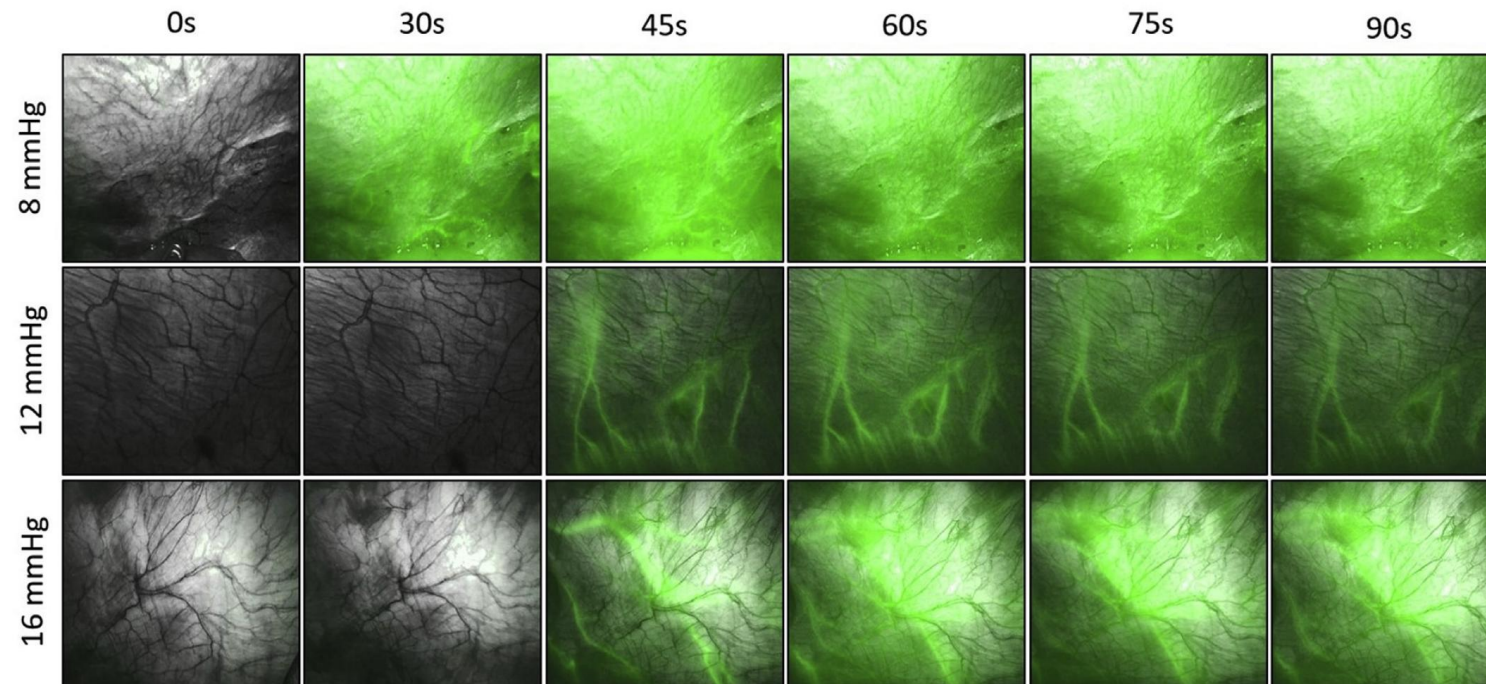


Visualising improved peritoneal perfusion at lower intra-abdominal pressure by fluorescent imaging during laparoscopic surgery: A randomised controlled study



Kim I. Albers^{a,b,*}, Fatih Polat^c, Tom Loonen^d, Leon J. Graat^e, Jan P. Mulier^f, Marc MJ. Snoeck^g, Ivo F. Panhuizen^g, Ad A. Vermulst^h, Gert-Jan Scheffer^b, Michiel C. Warlé^a

International Journal of Surgery 77 (2020) 8–13



Visualising improved peritoneal perfusion at lower intra-abdominal pressure by fluorescent imaging during laparoscopic surgery: A randomised controlled study



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Table 2
Study parameters.

	8 mmHg	12 mmHg	16 mmHg
Time to maximal fluorescent intensity (seconds)	43.7 ± 11.9 (34–66)	59.6 ± 15.2 (38–84)	57.3 ± 21.3 (29–90)
Maximal fluorescent intensity [–]	222 ± 25 (178–252)	172 ± 58 (93–254)	203 ± 47 (132–254)
Blood pressure (mmHg)	133/75 (94)	135/78 (97)	127/81 (96)
Heartrate (bpm)	66 ± 16	63 ± 15	63 ± 10
Patient temperature (°C)	35.8 ± 0.3	36.1 ± 0.2	36.2 ± 0.6
Rocuronium dose (mg/kg) ^a	0.78 ± 0.3	0.65 ± 0.1	0.74 ± 0.2



Detecção Clínica da HIA - SCA

- **Sinais clínicos**

- Sedação/Analgesia/IOT/ VM
- Exame físico ??
- Oliguria progressiva
- Piora da complacência respiratória
- Hipotensão e retenção periférica de fluidos



IS CLINICAL EXAMINATION AN ACCURATE INDICATOR OF RAISED INTRA-ABDOMINAL PRESSURE IN CRITICALLY INJURED PATIENTS?

Andrew W. Kirkpatrick, MD;*Frederick D. Brenneman, MD;† Richard F. McLean, MD;‡ Theodore Rapanos, BSc, MSc;§ Bernard R. Boulanger, MD¶

Table III

Comparison of the Findings From Clinical Examination of the Abdomen and Measured Bladder Pressure (Elevated Pressure Defined as > 15 mm Hg)

Clinical examination	Measured bladder pressure	
	> 15 mm Hg	≤ 15 mm Hg
Elevated	9	17
Normal	7	114

- Exame físico para PIA $>15\text{mmHg}$
- Sensibilidade 56%



***** Não deve ser utilizado para confirmar ou excluir a entidade!!!**



Pacientes que devem ser rotineiramente rastreados para SCA:

Table 2 Should patients be routinely screened for IAH and ACS?

Studies	Design	Quality	Consistency	Directness	<i>n</i>	Findings
4 burn [19, 26, 37, 38]	3 prospective, 1 retrospective	No serious limitations	Some inconsistency ^a	Direct	1,067	IAH incidence 36–70%, ACS incidence 1–20%
1 liver transplant [29]	Prospective	No serious limitations	Only one study	Direct	108	IAH incidence 32%
3 abdominal surgery [13, 20, 39]	Prospective	No serious limitations	No important inconsistency	Direct	424	IAH incidence 33–41%
11 trauma [4, 8, 10, 11, 21–23, 34, 40–42]	6 prospective, 5 retrospective	No serious limitations	Some inconsistency ^b	Direct	3,070	IAH incidence 2–50%, ACS incidence 0.5–36%
2 medical/surgical [2, 3]	Prospective	No serious limitations	No important inconsistency	Direct	322	IAH prevalence 32–54%, ACS prevalence 4–8%
3 pancreatitis [28, 43, 44]	1 prospective, 2 retrospective	No serious limitations	Some inconsistency ^a	Direct	361	IAH incidence 30–78%, ACS incidence 36%
2 pediatric [45, 46]	Prospective	No serious limitations	No important inconsistency	Direct	2,814	ACS incidence 0.6–0.9%

^a IAH/ACS definition variable

^b intervention thresholds and management variable



WSACS Guidelines

Cheatham, ICM 2006

Patient meets **ONE** of the following criteria
and has at least **TWO risk factors** for IAH:

1. New intensive care unit admission
2. Evidence of clinical deterioration or
new organ failure

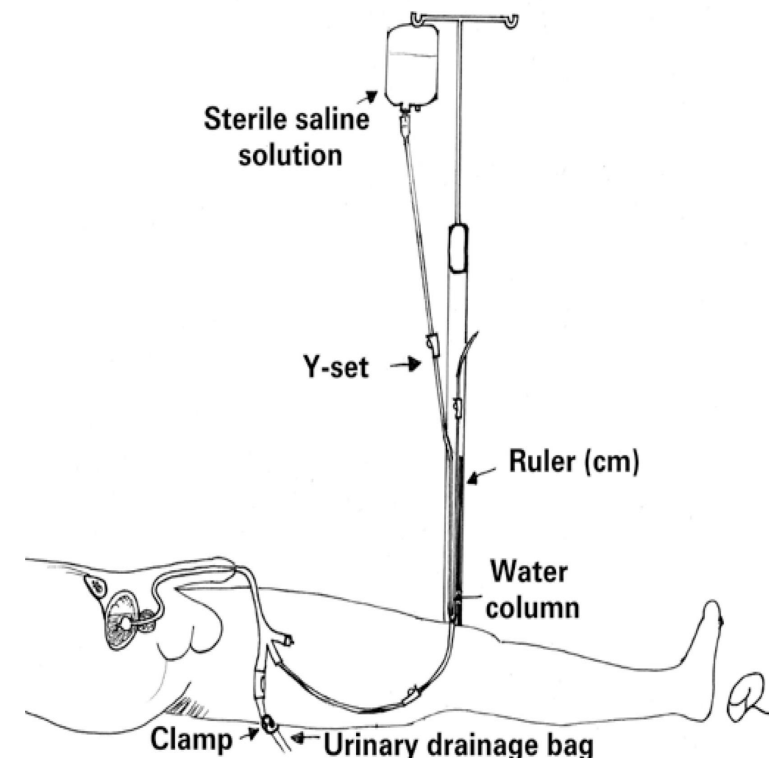
Risk Factors for IAH / ACS

1. Diminished abdominal wall compliance
 - Acute respiratory failure, especially with elevated intrathoracic pressure
 - Abdominal surgery with primary fascial closure
 - Major trauma / burns
 - Prone positioning
2. Increased intra-luminal contents
 - Gastroparesis
 - Ileus
 - Colonic pseudo-obstruction
3. Increased abdominal contents
 - Hemoperitoneum / pneumoperitoneum
 - Ascites / liver dysfunction
4. Capillary leak / fluid resuscitation
 - Acidosis ($\text{pH} < 7.2$)
 - Hypotension
 - Hypothermia (core temperature $< 33^{\circ}\text{C}$)
 - Polytransfusion (> 10 units of blood / 24 hrs)
 - Coagulopathy (platelets $< 55000 / \text{mm}^3$ OR activated partial thromboplastin time (APTT) > 2 times normal OR prothrombin time (PTT) $< 50\%$ OR International standardised ratio (INR) > 1.5)
 - Massive fluid resuscitation ($> 5 \text{ L} / 24 \text{ hours}$)
 - Oliguria
 - Sepsis
 - Major trauma / burns
 - Damage control laparotomy

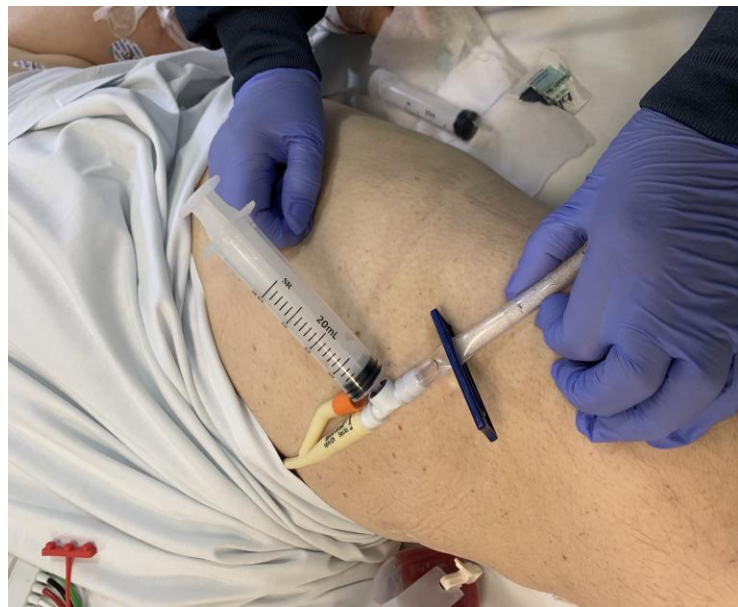


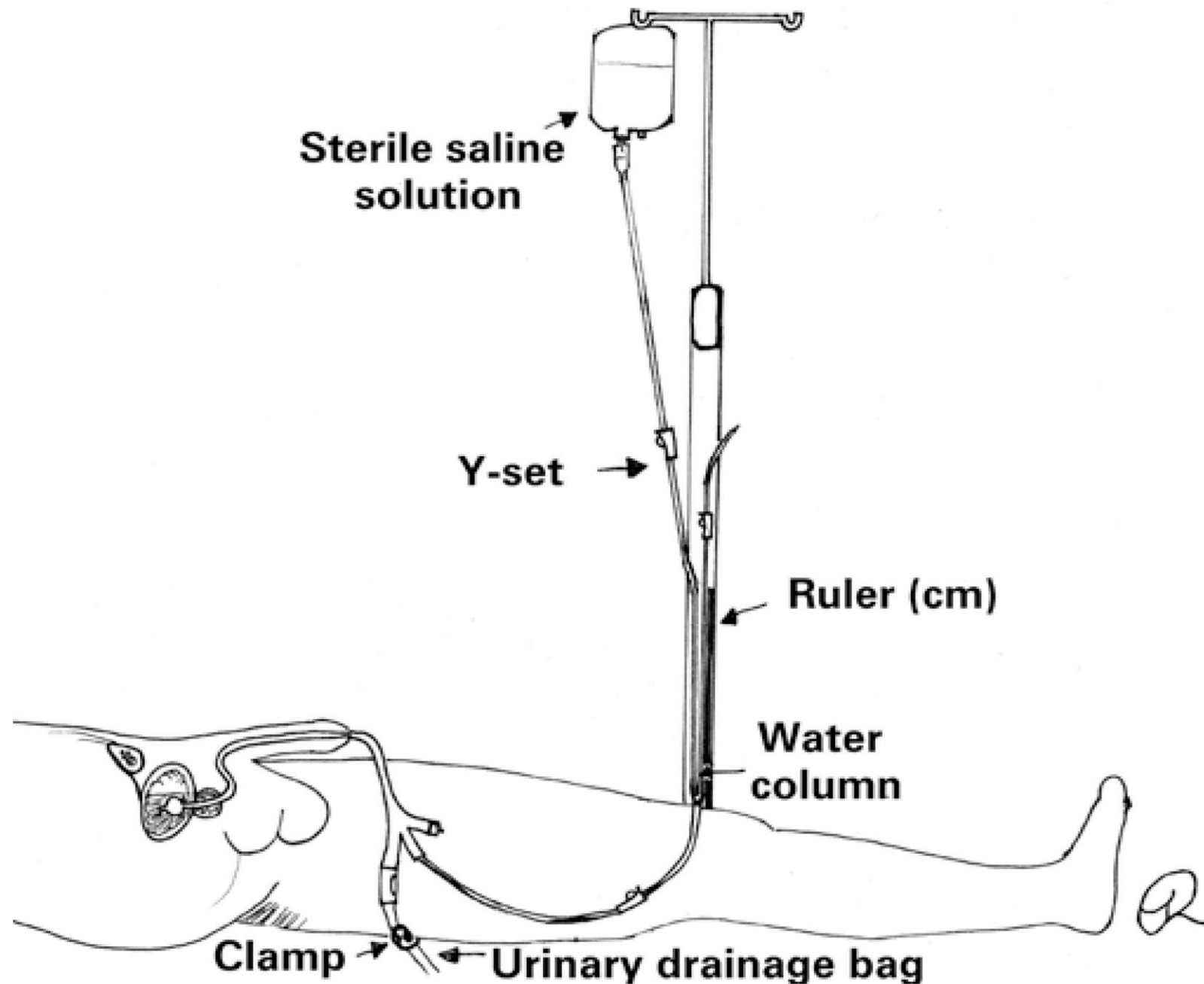
Como Aferir?

- **Fim da expiração**
- Supino
- Nivelar com Linha axilar média
- **Instilação de até 25ml de salina (Intravesical)**
- **Aferir 30-60 segundos após (relaxamento do detrusor)**
- Converter cmH₂O para mmHg (1 mmHg = 1.36 cm H₂O)







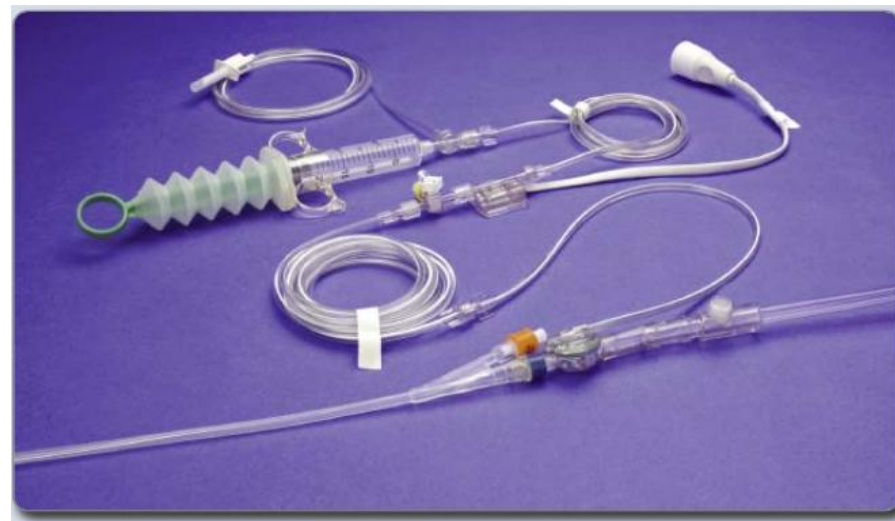
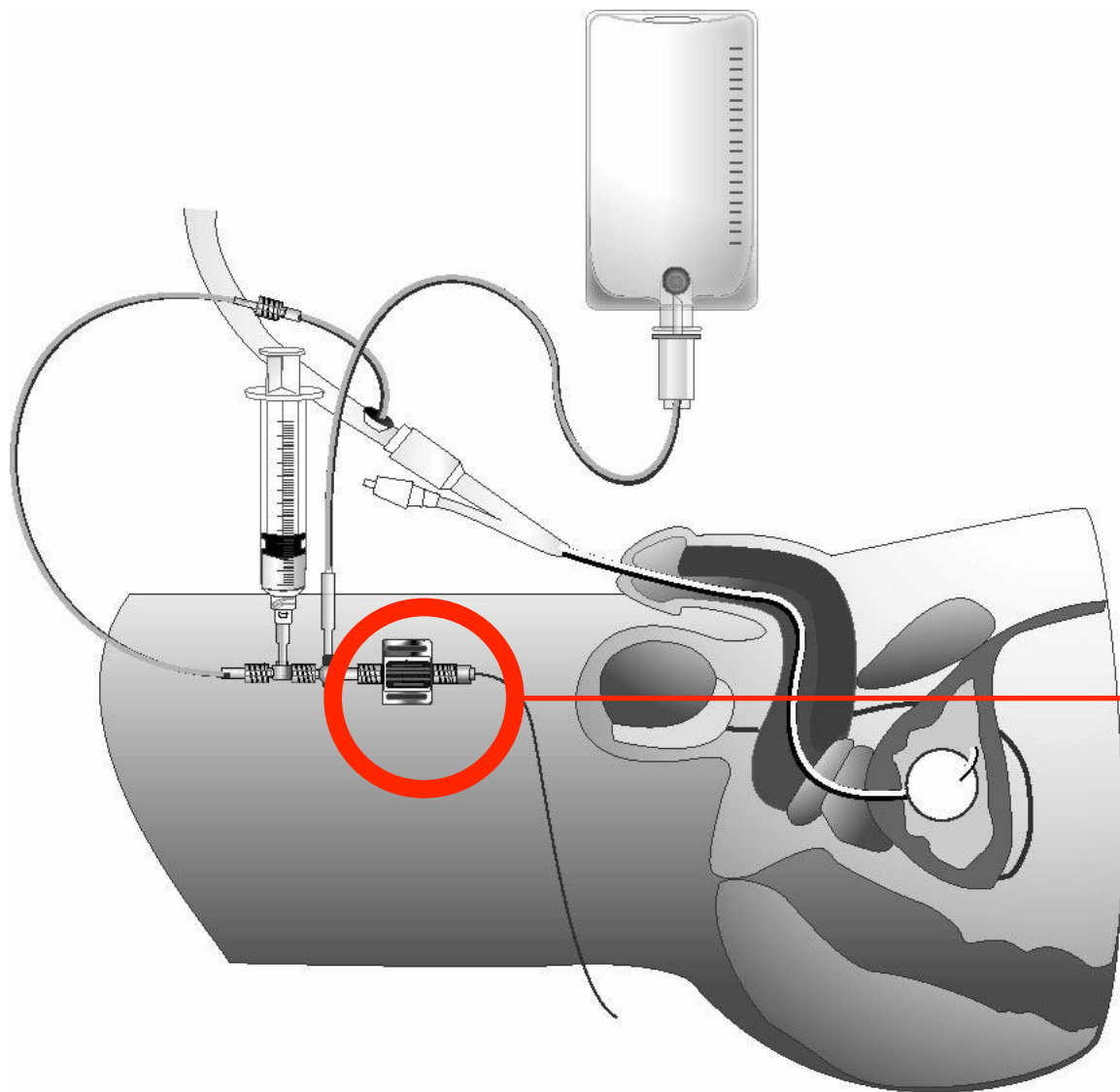


Dx = 2 aferições –
intervalo 1-6h
Monitoramento

*** bexiga neurogênica,
HPB, cistectomia –
impossibilidade aferição
= intra gástrica



Técnica de medida da PIA



Linha axilar média



Técnica de medida da PIA intravesical

- The current standard for monitoring abdominal pressures (*Consensus, World Congress ACS 2004*)
- Comparable to direct intraperitoneal pressure measurements, but is non-invasive (*Fusco 2001, Davis 2005, Risin 2006, Schachtrupp 2006*)
- More reliable and reproducible than clinical judgment (*Kirkpatrick, CJS 2000; Sugrue World J Surg 2002*)



Detecção da HIA - SCA

- **Exames de imagem**

- Radiografia

- USG

- TC



- Conteúdo luminal

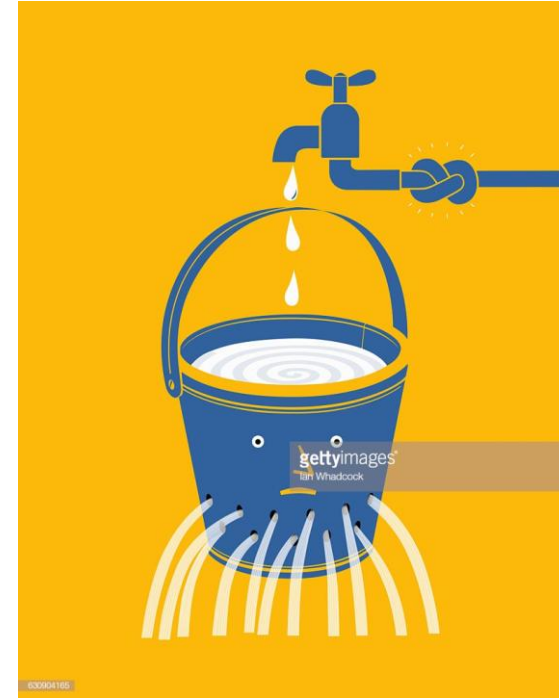
- Conteúdo extra-luminal

- Complacência ?



Tratamento Clínico:

- Aumento Complacência
- Redução do conteúdo luminal e extra luminal
- Retirada de líquidos
- PIA ↓, PPA ↑ , suporte as disfunções



Relação conteúdo-contenente



Tratamento

- Suporte Hemodinâmico : PPA > 60mmHg?, vasopressores precoce, avaliação DC
- Suporte Ventilatório

Table 3 Outcome Variables Stratified by Survival

	Survivors	Nonsurvivors	Significance of Multiple Logistic Regression
Lowest APP (mm Hg)	52 ± 17	39 ± 18	0.002
Lowest MAP (mm Hg)	74 ± 14	69 ± 14	0.05
Highest IAP (mm Hg)	29 ± 12	38 ± 14	0.21
Highest arterial lactate (mmol/L)	5.4 ± 2.2	8.0 ± 3.7	0.38
Highest base deficit	9.0 ± 7.0	13.1 ± 6.9	0.44
Lowest arterial pH	7.24 ± 0.10	7.14 ± 0.13	0.66
Lowest hourly urinary output (mL/h)	47 ± 48	44 ± 68	0.85



Posicionamento

Impact of the patient's body position on the intraabdominal workspace during laparoscopic surgery

Jan Paul J. Mulier • Bruno Dillemans •
Sebastiaan Van Cauwenberge

Table 1 Inflated volume and pressure for different body positions

	A Supine 	B Reverse trendelenburg 	C Beach chair 	D Horizontal, legs up 	E Trendelenburg 
IAP (mmHg)	14.0 ± 0.3	13.9 ± 0.5	14.0 ± 0.3	13.9 ± 0.2	14.1 ± 0.5
IAV (l)	3.22 ± 0.78	2.99 ± 0.69	3.76 ± 0.90	4.13 ± 0.84	4.14 ± 0.60
Minimum IAV (l)	2.00	1.99	2.39	3.05	3.33
Maximum IAV (l)	4.41	4.00	5.00	5.43	5.11
Patients with IAV < 3 l: n (%)	8 (40.0)	10 (50.0)	4 (20.0)	0 (0)	0 (0)

IAP intraabdominal pressure, IAV intraabdominal volume

Positions: A (supine: table horizontal with the legs flat), B (reverse Trendelenburg: table in 20° reverse Trendelenburg with the legs flat), C (beach chair: table in 20° reverse Trendelenburg with the legs elevated at 45°, D (horizontal, legs up: table horizontal with the legs elevated at 45°, E (Trendelenburg: table in 20° Trendelenburg with the legs flat)



Sedação / Analgesia / Bloqueio Neuromuscular

Redução dor, agitação, uso musculatura acessória e tônus

Intensive Care Med (2007) 33:1811–1814
DOI 10.1007/s00134-007-0758-0

Inneke De laet
Eric Hoste
Emmanuel Verholen
Jan J. De Waele

The effect of neuromuscular blockers in patients with intra-abdominal hypertension

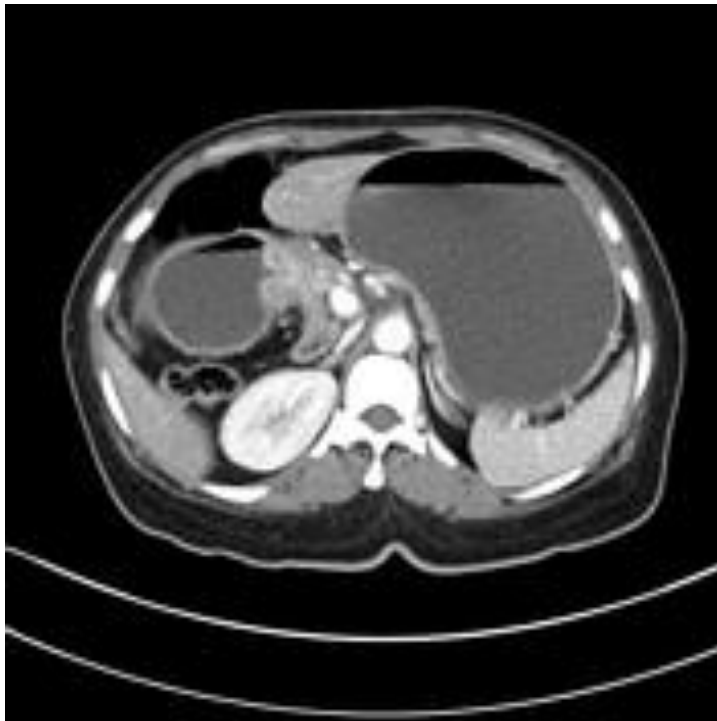
Table 1 Effect of bolus administration of cisatracurium on IAP, MAP, APP, CVP and heart rate

Parameter	Time T0	T15	T30	T60	T120	<i>p</i>
IAP	18 (16–20)	14 (12–16)*	14 (13–17)**	15 (14–17)	17 (14–19)	0.002
MAP	71 (64–76)	72 (62–77)	69 (64–77)	70 (63–77)	70 (62–88)	0.969
APP	53 (48–63)	58 (48–66)	59 (45–69)	56 (48–69)	54 (44–65)	0.674
CVP	18 (13–19)	16 (13–17)	15 (13–17)	15 (14–18)	18 (15–20)	0.146
HR	109 (93–113)	109 (94–115)	109 (96–116)	105 (94–111)	104 (99–110)	0.280

1 dose cisatracúrio = redução 4mmHg



Descompressão Gástrica / Colon / Retal/ Enterocinético



→ Conteúdo intra luminal: procinéticos, laxativos, SNG, sonda retal, suspensão dieta



Paracentese



Manipulation of Ascitic Fluid Pressure in Cirrhotics to Optimize Hemodynamic and Renal Function

TABLE 4. Renal Function Parameters Pre- and Postparacentesis

	Preparacentesis (51 studies) Mean $\pm$ SD		Postparacentesis (51 studies) Mean $\pm$ SD	
IAP (cm H ₂ O)	33.47 $\pm$	9.12	19.06 $\pm$	4.37*
Hematocrit (%)	32.0 $\pm$	2.22	32.33 $\pm$	1.94‡
BUN (mg/dl)	27.04 $\pm$	8.99	25.3 $\pm$	8.17*
Serum Creatinine (mg/dl)	1.37 $\pm$	0.49	1.32 $\pm$	0.58*
Serum Osmolality (mOsm)	289.0 $\pm$	9.77	287.47 $\pm$	8.14‡
Serum Sodium (mEq/L)	135.27 $\pm$	4.54	135.14 $\pm$	3.88‡
Urine Osmolality (mOsm)	447.51 $\pm$	130.91	495.63 $\pm$	152.7‡
Urine Volume (cc/hour)	46.74 $\pm$	26.46	54.95 $\pm$	24.52†
Urine Sodium (mEq/L)	51.18 $\pm$	33.41	54.27 $\pm$	29.2‡
Urine Creatinine (mg/dl)	85.47 $\pm$	54.15	97.69 $\pm$	56.17*
Ccr (cc/min)	46.22 $\pm$	18.2	63.73 $\pm$	20.9*
Cosm (cc/min)	71.40 $\pm$	35.39	94.9 $\pm$	46.2*
CH ₂ O (cc/min)	0.71 $\pm$	0.26	0.67 $\pm$	0.36‡
FeNa (%)	0.72 $\pm$	0.48	0.61 $\pm$	0.37‡
RFI (%)	0.98 $\pm$	0.66	0.82 $\pm$	51.0‡

* P < 0.001. †P < 0.01. ‡Not significant.





Percutaneous Catheter Decompression in the Treatment of Elevated Intraabdominal Pressure

Michael L. Cheatham, MD; and Karen Safcsak, RN

- Drenagem percutânea evitou cirurgia em 81% dos casos
- Drenagem percutânea eficaz >> Mais de 1L ou queda de 9mmHg da PIA

Percutânea = Cirurgia ?

Table 2—Comparative Physiology of Percutaneous and Open Abdominal Decompression Cohorts

Physiologic Variable	Percutaneous Decompression	Open Abdomen	P Value
Δ IAP, mm Hg	-7 ± 6	-12 ± 14	.30
Predecompression APP, mm Hg	55 ± 16	48 ± 18	.21
Postdecompression APP, mm Hg	64 ± 13	65 ± 17	.96
Δ APP, mm Hg	10 ± 15	15 ± 26	.54
Predecompression UOP, mL/h	43 ± 52	47 ± 48	.80
Postdecompression UOP, mL/h	75 ± 85	74 ± 68	.98



Balanço Hídrico

Risk factors for intra-abdominal hypertension and abdominal compartment syndrome among adult intensive care unit patients: a systematic review and meta-analysis

Jessalyn K Holodinsky¹, Derek J Roberts^{2*}, Chad G Ball³, Annika Reintam Blaser⁴, Joel Starkopf⁴, David A Zygun⁵, Henry Thomas Stelfox⁶, Manu L Malbrain⁷, Roman C Jaeschke⁸ and Andrew W Kirkpatrick⁹



Holodinsky et al. *Critical Care* 2013, **17**:R249
<http://ccforum.com/content/17/5/R249>

Table 4 Significant risk factors for intra-abdominal hypertension among intensive care unit patients, including pooled estimates where appropriate

Risk factor	Systematic review and/or meta-analysis					
	Number		Patient population	Odds ratio (95% CI)	Heterogeneity	
	Studies	Patients			I ² , %	P value
Crystalloid resuscitation						
Pre-ICU crystalloid [17] ^{†,‡}	1	81	Trauma patients	1.40 (1.00 to 1.96)		
Fluid balance [9] ^{†,‡}	1	563	Mixed ICU patients	5.22 (2.03 to 7.45)		
24hr fluid balance [10] ^{†,‡}	1	58	Pancreatitis patients	1.004 (1.001 to 1.006)*		
Fluid collections [10] [†]	1	58	Pancreatitis patients	2015 (1.298 to 3.129)*		



Nutrição

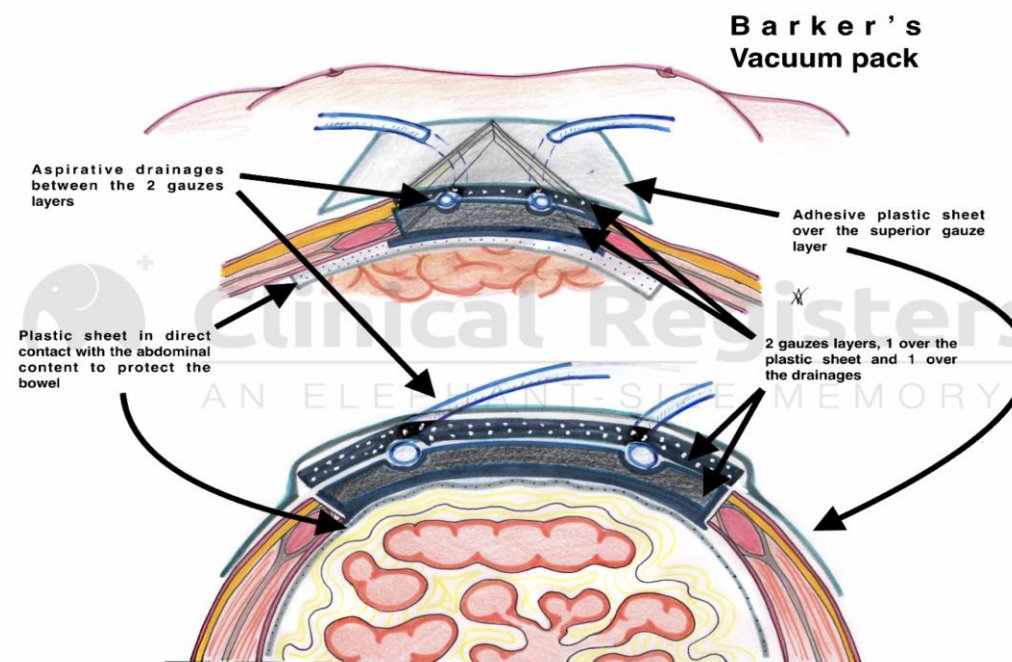
- Hipercatabolismo
- Se PIA controlada → nutrição enteral trófica
- HIA com progressão para SCA – suspensão de dieta, avaliar NPP a depender do quadro clínico



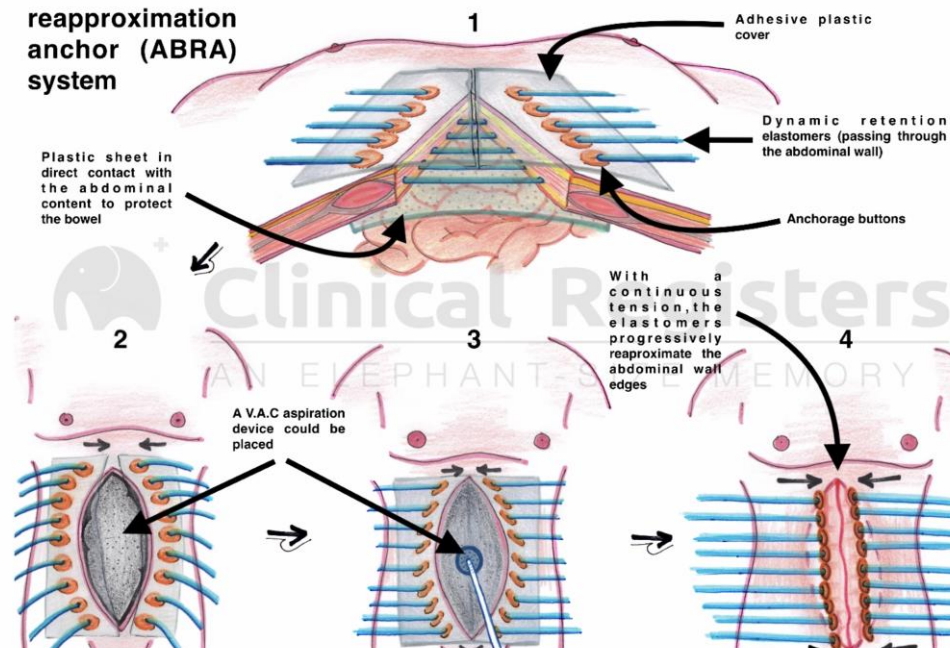
Tratamento Cirúrgico

- Prevenção! (hérnias incisionais gigantes, transplante órgãos, cirurgias redutoras)
- Coagulopatias, Volume, tensão, Piora ao fechar >> Não Fechar ?
- Falha do Tratamento Clínico (Pia >25?)
- Evolução da(s) Disfunções associadas a HIA.
- Vácuo? Abthera?





Abdominal reapproximation anchor (ABRA) system

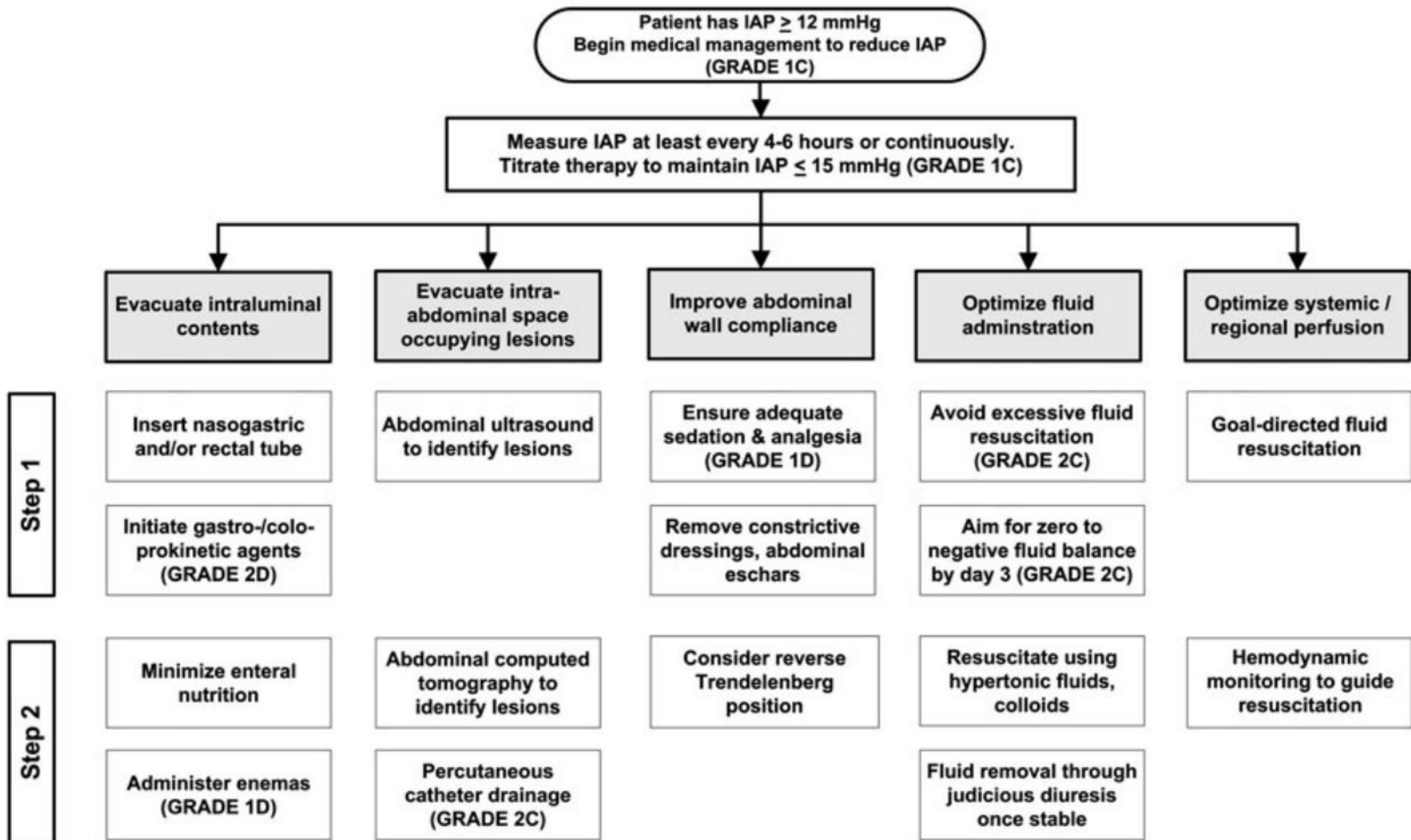


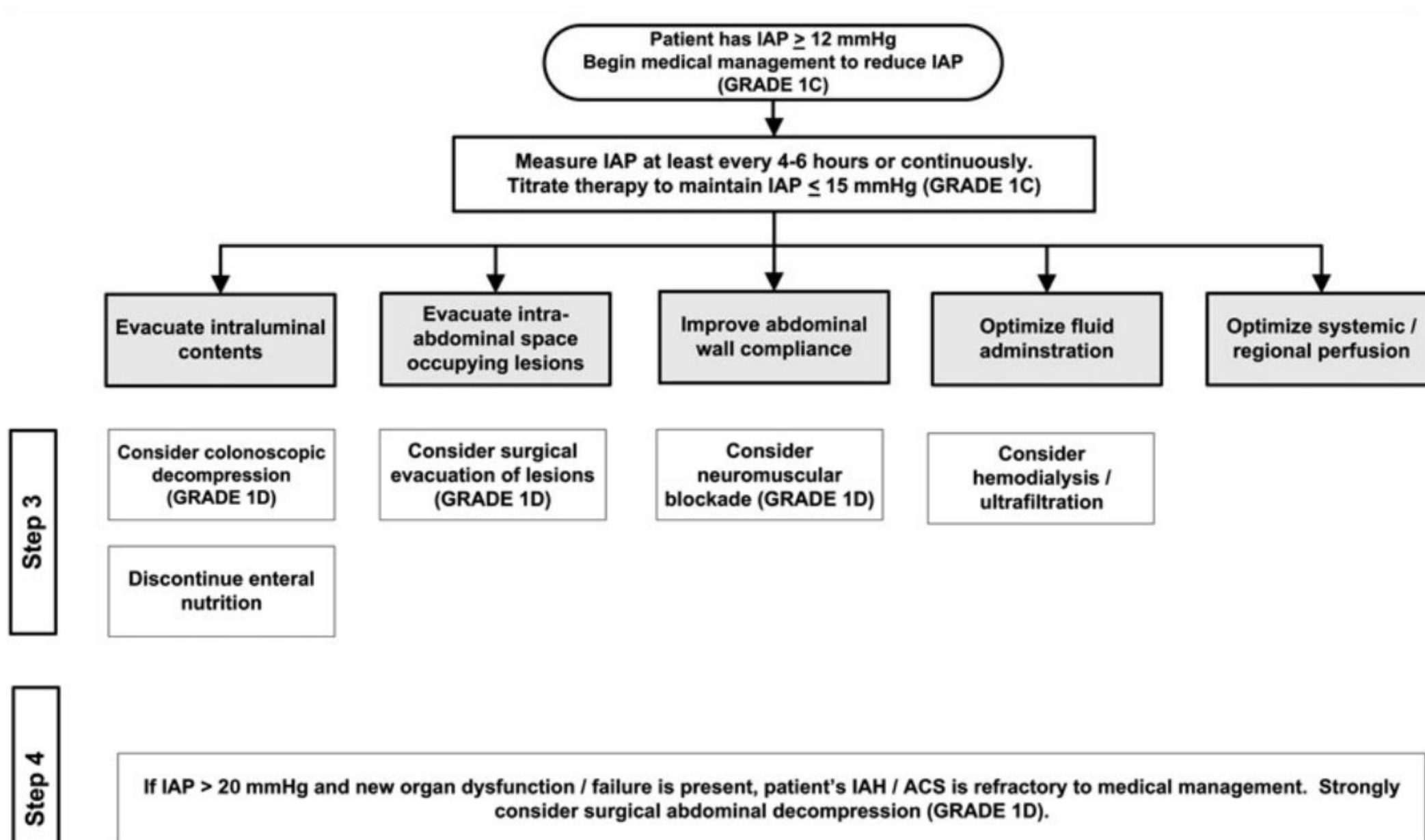


Intra abdominal hypertension and the abdominal compartment syndrome

Paul Hayden







Tratamento da SCA

Reduzir
conteúdo
Luminal

Reduzir
conteúdo
Extra-Luminal

Complacência
Abdominal

Fluidos /
Hemodinâmica

- Sonda gástrica
- Sonda Retal
- Pró cinéticos
- Jejum
- Enemas
- Descompressão endoscópica

- Imagem abdome
- Paracentese
- Radio intervenção
- Cirurgia

- Analgesia
- Sedação
- Remover faixas ou restrições
- Bloqueio neuromuscular

- PPA > 60 mmHg
- Balanço negativo
- Diuréticos
- Avaliar TSR

PIA > 25 ou Refratariedade >> Tratamento Cirúrgico



Conclusão

- Síndrome de alta morbimortalidade
- Diagnóstico clínico/exames complementares falho >> Alta suspeição
- Terapia clínica de agressiva de primeira linha
- Tratamento cirúrgico preventivo e resolutivo! (precoce?)



Obrigada!

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